

XAI Lecture 23

Understanding and Reasoning in Large Language Models

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eXplainable
Artificial Intelligence

Disclaimer ---

This course is based on

Explainable Artificial Intelligence

From Simple Predictors to Complex Generative Models

Spring 2023, Harvard University

<https://interpretable-ml-class.github.io/>

Chain-of-Thought Prompting Elicits Reasoning in Large Language Models

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(Wei et al. 2022)

Challenges in Large language models (LLMs)

- LLMs have limitations in their ability to reason and understand the context of a situation.
- Scaling up model size alone has not proved sufficient for achieving high performance on challenging tasks, such as *arithmetic*, *commonsense*, and *symbolic reasoning*.

Previous work

- LLMs have the ability to generate natural language intermediate steps by training from scratch (Ling et al. 2017) or finetuning a pretrained model (Cobbe et al. 2021).
 - ▶ It is costly to create a large set of high quality rationales.
- LLMs offer the exciting prospect of in-context few-shot learning via *prompting* (Brown et al. 2020).
 - ▶ Works poorly on tasks that require reasoning abilities.
 - ▶ Often does not improve substantially with increasing model scale.

Contribution ---

- Combining few-shot prompting with explicit intermediate reasoning steps, avoiding the limitations of both prior approaches.
 - ▶ Prompts consist of triples $\langle \text{input}, \text{chain of thought}, \text{output} \rangle$
 - ▶ Chain of thought (CoT): series of intermediate natural language reasoning steps leading to the final answer.
 - ▶ No large training dataset required — a single model checkpoint can perform many tasks without loss of generality.
- Showing sufficiently large LMs can generate CoT if demonstrations are provided as few-shot prompting.
- Empirical evaluations on **arithmetic, commonsense, and symbolic reasoning** benchmarks, showing that CoT prompting outperforms standard prompting.

CoT Prompting ---

When solving a complex reasoning task, we naturally tend to decompose it into intermediate steps and solve each one before giving the final answer.

Example

"After Jane gives 2 flowers to her mom she has 10 ... then after she gives 3 to her dad she will have 7 ... so the answer is 7."

Goal of the paper: to endow LLMs with the ability to generate a similar CoT—a coherent series of intermediate reasoning steps that lead to the final answer for a problem.

Standard Prompting vs. CoT Prompting ---

Standard Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The answer is 27. 

Chain-of-Thought Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had $23 - 20 = 3$. They bought 6 more apples, so they have $3 + 6 = 9$. The answer is 9. 



More examples of CoT Prompting ---

Math Word Problems (free response)

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Math Word Problems (multiple choice)

Q: How many keystrokes are needed to type the numbers from 1 to 500?
Answer Choices: (a) 1156 (b) 1392 (c) 1480 (d) 1562 (e) 1788

A: There are 9 one-digit numbers from 1 to 9. There are 90 two-digit numbers from 10 to 99. There are 401 three-digit numbers from 100 to 500. $9 + 90(2) + 401(3) = 1392$. The answer is (b).

CSQA (commonsense)

Q: Sammy wanted to go to where the people were. Where might he go?
Options: (a) race track (b) populated areas (c) desert (d) apartment (e) roadblock

A: The answer must be a place with a lot of people. Race tracks, desert, apartments, and roadblocks don't have a lot of people, but populated areas do. So the answer is (b).

StrategyQA

Q: Yes or no: Would a pear sink in water?

A: The density of a pear is about 0.6 g/cm^3 , which is less than water. Thus, a pear would float. So the answer is no.

Date Understanding

Q: The concert was scheduled to be on 06/01/1943, but was delayed by one day to today. What is the date 10 days ago in MM/DD/YYYY?

A: One day after 06/01/1943 is 06/02/1943, so today is 06/02/1943. 10 days before today is 05/23/1943. So the answer is 05/23/1943.

Sports Understanding

Q: Is the following sentence plausible? "Joao Moutinho caught the screen pass in the NFC championship."

A: Joao Moutinho is a soccer player. The NFC championship is part of American football, not soccer. So the answer is no.

SayCan (Instructing a robot)

Human: How would you bring me something that isn't a fruit?

Explanation: the user wants something to eat that isn't a fruit. An energy bar is not a fruit, so I will bring the user an energy bar.

Plan: 1. find(energy bar) 2. pick(energy bar) 3. find(user) 4. put(energy bar) 5. done().

Last Letter Concatenation

Q: Take the last letters of the words in "Lady Gaga" and concatenate them.

A: The last letter of "Lady" is "y". The last letter of "Gaga" is "a". Concatenating them is "ya". So the answer is ya.

Coin Flip (state tracking)

Q: A coin is heads up. Maybelle flips the coin. Shalonda does not flip the coin. Is the coin still heads up?

A: The coin was flipped by Maybelle. So the coin was flipped 1 time, which is an odd number. The coin started heads up, so after an odd number of flips, it will be tails up. So the answer is no.

Benefits of Chain-of-Thought ---

- ➊ Allows models to decompose multi-step problems into intermediate steps so additional computation can be allocated to problems that require more reasoning steps.
- ➋ Provides an interpretable window into the behavior of the model, suggesting how it might have arrived at a particular answer and providing opportunities to debug where the reasoning path went wrong (although fully characterizing a model's computations that support an answer remains an open question).
- ➌ Potentially applicable to any task that requires a multi-step approach.
- ➍ Can be readily elicited in sufficiently large off-the-shelf LLMs simply by including examples of CoT sequences while prompting.

General Experimental Setup ---

Objective: to observe the utility of CoT prompting for *arithmetic reasoning*, *commonsense reasoning*, and *symbolic reasoning*.

Prompts:

- Baseline: standard few-shot prompting (Brown et al. 2020), where exemplars are formatted as questions and answers.
- Chain-of-thought: augment each exemplar in few-shot prompting with a CoT for an associated answer.

Models:

- GPT-3 (Brown et al. 2020).
- LaMDA (Thoppilan et al. 2022).
- PaLM (Chowdhery et al. 2022).
- UL2 20B (Tay et al. 2022).
- Codex (Chen et al. 2021).

Arithmetic Reasoning: Experimental Setup ---

Benchmarks:

- **GSM8K** (Cobbe et al. 2021)
- **SVAMP** (Patel et al. 2021)
- **ASDiv** (Miao et al. 2020)
- **AQuA** (Ling et al. 2017)
- **MAWPS** (Koncel-Kedziorski et al. 2016)

Math Word Problems (free response)

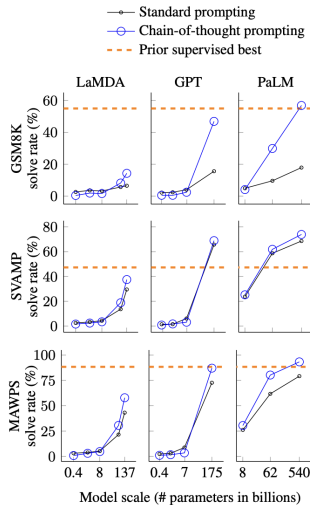
Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

🍷 Arithmetic Reasoning: Results ---

Three key takeaways:

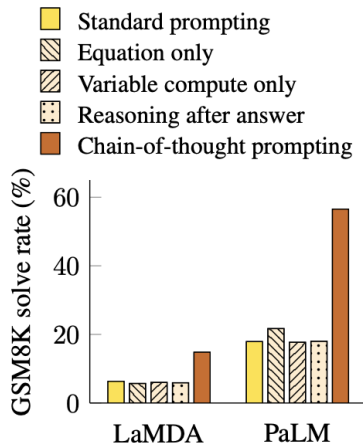
- CoT prompting is an emergent ability of **model scale**.
- Larger performance gains for **more-complicated problems**.
- GPT-3 175B and PaLM 540B **compare favorably to prior state of the art**.



🍷 Ablation Study ---

A study with three variations:

- **Equation only:** only the math equation, no natural language.
 - ▶ Still fails on hard problems — semantics matter.
- **Variable compute only:** dots (...) matching CoT length (same tokens, no reasoning).
 - ▶ No improvement — extra compute alone is not enough.
- **Chain of thought after answer:** reasoning appears *after* the answer.
 - ▶ Also fails — sequential reasoning *before* the answer is essential.



🍷 Robustness of Chain of Thought ---

Chain of thought is robust to:

- 1 Independently-written chains of thought:
 - 1 Different annotators.
 - 2 Annotators without ML background.
- 2 Different types, ordering, and number of exemplars.
- 3 Different language models (LaMDA, GPT-3, PaLM).

However, prompt engineering improved performance significantly in many cases.

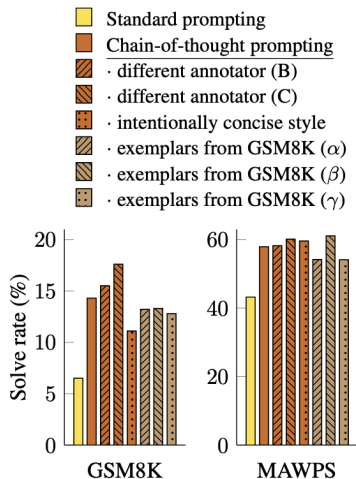


Figure shows results for LaMDA 137B.

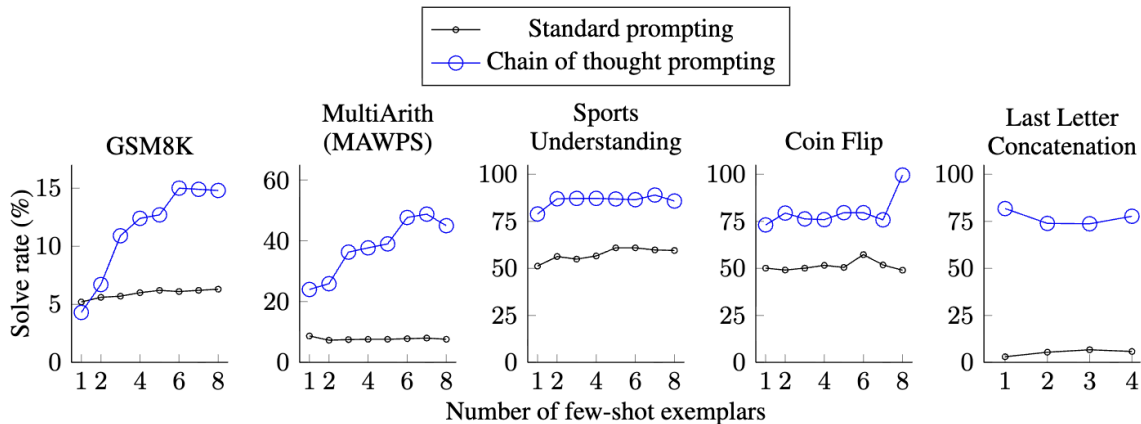


Figure shows results for LaMDA 137B.

Commonsense Reasoning: Experimental Setup ---

Benchmarks:

- **CSQA** (Talmor et al. 2019).
- **StrategyQA** (Geva et al. 2021).
- **Date Understanding** (set from the BIG-bench (Srivastava et al. 2022)).
- **Sports Understanding** (set from the BIG-bench (Srivastava et al. 2022)).
- **SayCan** (Ahn et al. 2022).

Date Understanding

Q: The concert was scheduled to be on 06/01/1943, but was delayed by one day to today. What is the date 10 days ago in MM/DD/YYYY?

A: One day after 06/01/1943 is 06/02/1943, so today is 06/02/1943. 10 days before today is 05/23/1943. So the answer is 05/23/1943.

🍷 Commonsense Reasoning: Results ---

- CoT gains scale with model size: PaLM 540B achieves best results across all benchmarks.
- New SOTA on StrategyQA: 75.6% vs. prior best 69.4%.
- Surpasses human baseline on Sports Understanding: 95.4% vs. 84%.
- Minimal gains on CSQA: CoT benefits complex multi-step reasoning more than factual recall.



Commonsense Reasoning: Results ---

Model		CSQA		StrategyQA		Date		Sports		SayCan	
		standard	CoT	standard	CoT	standard	CoT	standard	CoT	standard	CoT
UL2	20B	34.2	51.4	59.0	53.3	13.5	14.0	57.9	65.3	20.0	41.7
LaMDA	420M	20.1	19.2	46.4	24.9	1.9	1.6	50.0	49.7	7.5	7.5
	2B	20.2	19.6	52.6	45.2	8.0	6.8	49.3	57.5	8.3	8.3
	8B	19.0	20.3	54.1	46.8	9.5	5.4	50.0	52.1	28.3	33.3
	68B	37.0	44.1	59.6	62.2	15.5	18.6	55.2	77.5	35.0	42.5
	137B	53.6	57.9	62.4	65.4	21.5	26.8	59.5	85.8	43.3	46.6
GPT	350M	14.7	15.2	20.6	0.9	4.3	0.9	33.8	41.6	12.5	0.8
	1.3B	12.0	19.2	45.8	35.7	4.0	1.4	0.0	26.9	20.8	9.2
	6.7B	19.0	24.0	53.6	50.0	8.9	4.9	0.0	4.4	17.5	35.0
	175B	79.5	73.5	65.9	65.4	43.8	52.1	69.6	82.4	81.7	87.5
Codex	-	82.3	77.9	67.1	73.2	49.0	64.8	71.7	98.5	85.8	88.3
PaLM	8B	19.8	24.9	55.6	53.5	12.9	13.1	55.1	75.2	34.2	40.0
	62B	65.4	68.1	58.4	63.4	29.8	44.7	72.1	93.6	65.8	70.0
	540B	78.1	79.9	68.6	77.8	49.0	65.3	80.5	95.4	80.8	91.7

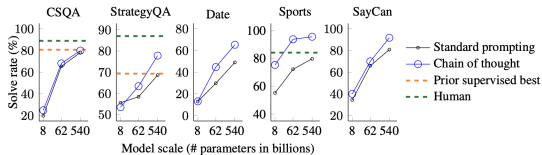


Figure above shows results for PaLM.

Symbolic Reasoning: Experimental Setup ---

Tasks:

- **Last Letter Concatenation:** concatenate the last letters of words in a name (e.g., “Lady Gaga” $\rightarrow$ “ya”).
- **Coin Flip:** track whether a coin is heads or tails after a series of flips.

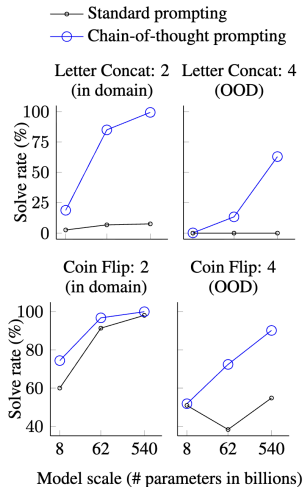
Evaluation:

- **In-domain (ID):** test examples of the same length as the few-shot demonstrations.
- **Out-of-domain (OOD):** test examples with more steps than shown in the prompt — evaluates length generalization.

Symbolic Reasoning: Results ---

- CoT enables **length generalization**: solves longer sequences than those shown in few-shot examples.
- Standard prompting fails completely on OOD symbolic tasks — CoT succeeds.
- PaLM 540B achieves near-perfect accuracy on ID tasks.
- Again, emergent ability: benefits only appear at models with 100B+ parameters.

🍷 Symbolic Reasoning: Results ---



Model		Last Letter Concatenation						Coin Flip (state tracking)					
		2		OOD: 3		OOD: 4		2		OOD: 3		OOD: 4	
		standard	CoT	standard	CoT	standard	CoT	standard	CoT	standard	CoT	standard	CoT
UL2	20B	0.6	18.8	0.0	0.2	0.0	0.0	70.4	67.1	51.6	52.2	48.7	50.4
LaMDA	420M	0.3	1.6	0.0	0.0	0.0	0.0	52.9	49.6	50.0	50.5	49.5	49.1
	2B	2.3	6.0	0.0	0.0	0.0	0.0	54.9	55.3	47.4	48.7	49.8	50.2
	8B	1.5	11.5	0.0	0.0	0.0	0.0	52.9	55.5	48.2	49.6	51.2	50.6
	68B	4.4	52.0	0.0	0.8	0.0	2.5	56.2	83.2	50.4	69.1	50.9	59.6
	137B	5.8	77.5	0.0	34.4	0.0	13.5	49.0	99.6	50.7	91.0	49.1	74.5
PaLM	8B	2.6	18.8	0.0	0.0	0.0	0.2	60.0	74.4	47.3	57.1	50.9	51.8
	62B	6.8	85.0	0.0	59.6	0.0	13.4	91.4	96.8	43.9	91.0	38.3	72.4
	540B	7.6	99.4	0.2	94.8	0.0	63.0	98.1	100.0	49.3	98.6	54.8	90.2

Figure above shows results for PaLM.

Conclusion ---

Through experiments on arithmetic, symbolic, and commonsense reasoning, we find that chain-of-thought reasoning is an emergent property of model scale that allows sufficiently large language models to perform reasoning tasks that otherwise have flat scaling curves.

Limitations & Discussion ---

Limitations

- CoT emulates human reasoning but does not prove the network actually "reasons".
- Emergence only at large model scales makes deployment costly.
- Generated chains of thought can be incorrect even when the final answer is right.

Discussion Questions

- What reasoning tasks may not be well-suited to chain-of-thought?
- What other prompting methods might expand LM capabilities?
- Does chain-of-thought truly improve interpretability?

Can language models learn from explanations in context?

Andrew K. Lampinen, Ishita Dasgupta, Stephanie C. Y. Chan

Kory Mathewson, Michael Henry Tessler, Antonia Creswell

James L. McClelland, Jane X. Wang, Felix Hill

DeepMind
London, UK

(Lampinen et al. 2022)



Introduction ---

- Language models (LMs) have been found to be able to perform new tasks by adapting to a few in-context examples (Brown et al. 2020).
 - ▶ In-context example: an example (question + right answer) for a specific task prompt.
 - ▶ Few-shot: a few examples are given to guide the model.
- **Key question:** Could including few-shot *explanations* of the answers in these examples improve LM performance?

🍷 Few-shot + Explanation ---

Task instruction { Answer these questions by identifying whether the second sentence is an appropriate paraphrase of the first, metaphorical sentence.

Few-shot example #1 { Q: David's eyes were like daggers at Paul when Paul invited his new girlfriend to dance. <- -> David had two daggers when Paul invited his new girlfriend to dance.
choice: True
choice: False
A: False

Answer explanation { Explanation: David's eyes were not literally daggers, it is a metaphor used to imply that David was glaring fiercely at Paul.

4 more examples + explanations

Target question { Q: Our whole life we swim against the waves towards the green light of happiness. <- -> Our whole life we try to reach happiness.
choice: True
choice: False
A:

Related Work ---

In-context and prompt-based learning:

- Min et al., 2022: Find that ground truth labels don't have a large effect on model performance, and identify other aspects (label space / distribution) that drive performance.
- Webson and Pavlick, 2021: Find limitation in models' ability to truly understand the meaning of their prompts.

Prompting with explicit instructions:

- Liu et al., 2021: Prompting with explicit instructions or task descriptions helps LMs adapt to a task.
- Wei et al, 2022: Breaking down the steps of a reasoning process for LMs improve few-shot performance.

This paper focuses specifically on the effect of **post-answer** explanations — distinct from chain-of-thought which places reasoning *before* the answer.

Research Questions ---

- Can language models benefit from explanations when learning from examples in-context?
- Do few-shot explanations help the model “understand” the task better?
- What kind of in-context learning abilities do LMs exhibit more generally?

Contributions ---

- Annotated **40 challenging tasks** from BIG-Bench with human-written explanations.
- Evaluated the effect of **post-answer explanations** (contrast to Wei et al.'s pre-answer chains).

Key findings

- Explanations improve LLM performance compared to matched control conditions.
- Explanations tuned on a small validation set yield even larger gains.

Methods: Data & Model ---

Data: 40 tasks from BIG-Bench, including:

- Inferring goals from actions.
- Reasoning about mathematical induction.
- Reasoning about causality.
- Inferring assumptions behind statements.

Model: Decoder-only Transformer LMs (Gopher family: 1B, 7B, 280B parameters), evaluated with the same context window and training data.

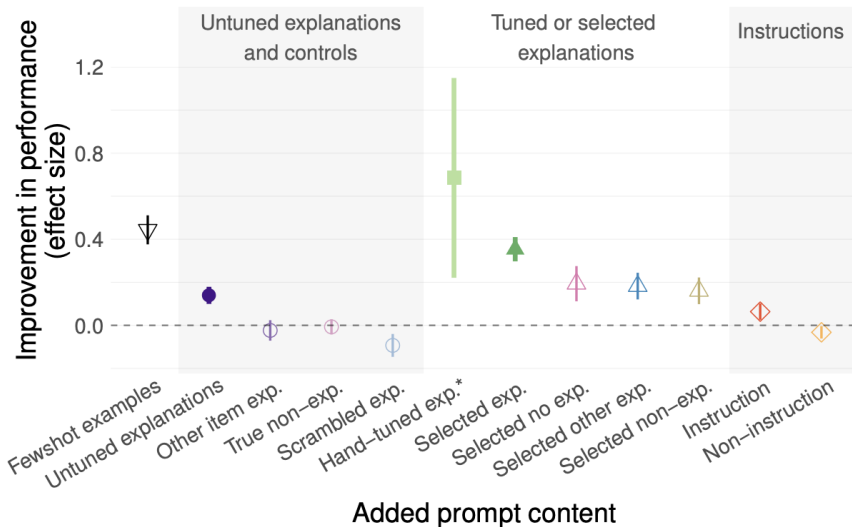
Methods: Explanation Annotation ---

- One author annotated **15 examples per task** with human explanations (restricted to multiple-choice tasks).
- Three **control conditions** to isolate the causal effect of explanations:
 - ▶ *True non-explanations*: relevant but non-explanatory text.
 - ▶ *Other item explanations*: explanations drawn from a different task example.
 - ▶ *Scrambled explanations*: words of the explanation randomly reordered.
- **Explanation tuning**: greedily select best 5-shot prompt; also hand-tune explanation wording.

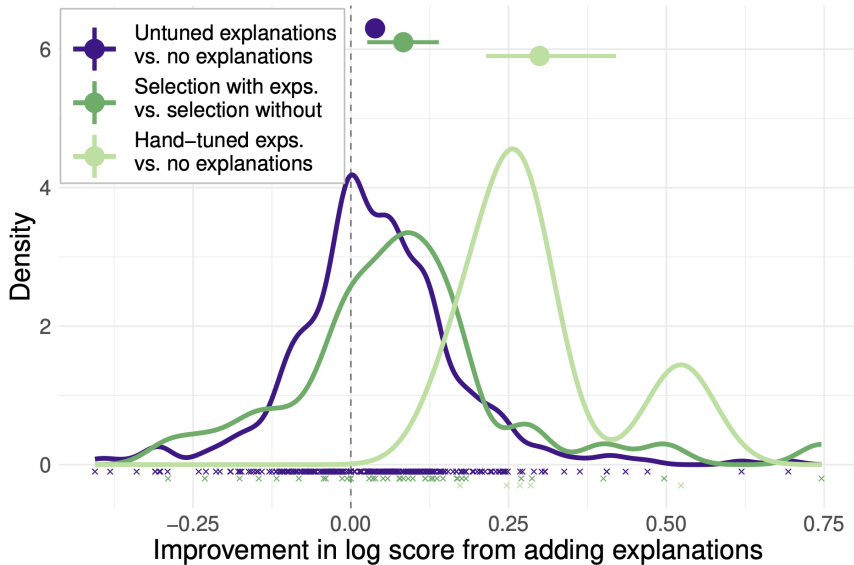
Methods: Evaluation ---

- Results modeled with **hierarchical logistic regression** to account for:
 - ▶ Task difficulty.
 - ▶ Shared content between prompts.
- Each effect estimated at the appropriate level of the hierarchy, quantifying the **unique added contribution** of each prompt component.

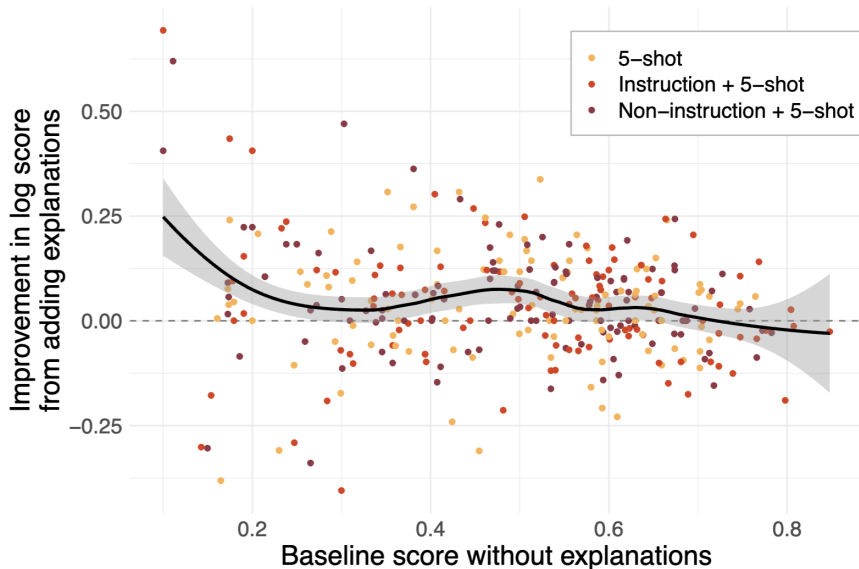
🍷 Effect Size (Gopher 280B) ---



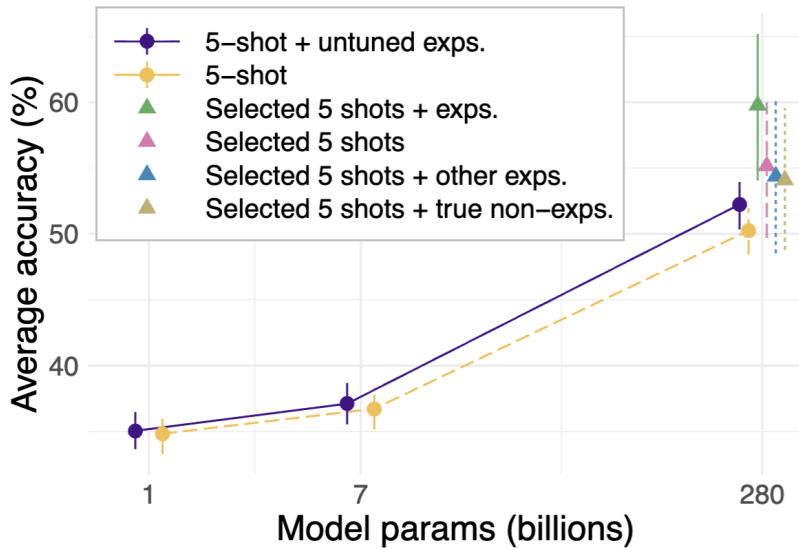
🍷 Selection vs. Untuned Explanations ---



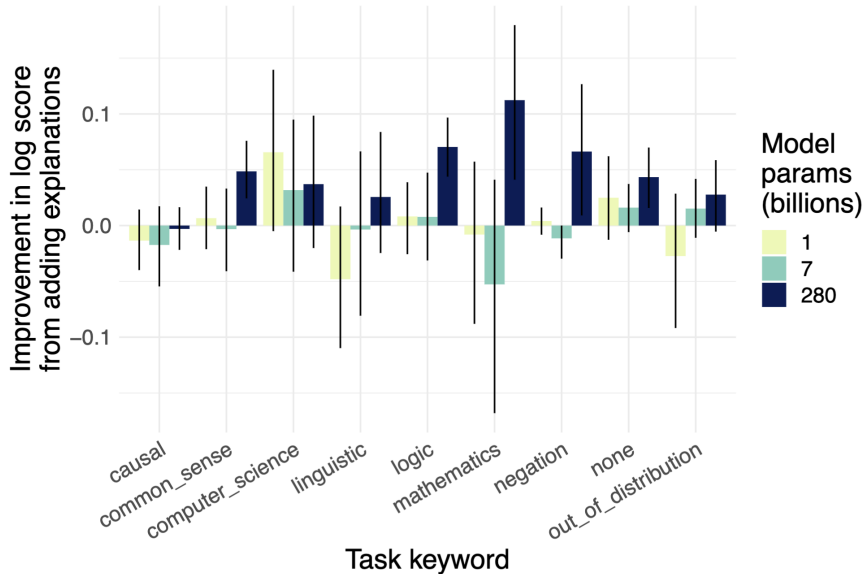
🍷 When Do Explanations Help? ---



🍷 Is Scale Necessary? ---



🍷 Not All Tasks Are Created Equal ---



Open questions

- Can explanations improve few-shot learning? Yes, with sufficient scale and quality.
- Why are post-answer explanations interesting? They decouple reasoning from generation and have no test-time cost.
- What do results imply about in-context learning? LMs can extract semantic content from explanations, not just surface structure.
- How do explanations relate to instructions? They appear complementary.
- How does this relate to human language processing? LMs differ — they lack broader world context.

Conclusion ---

- Including explanations with examples in a few-shot prompt can improve in-context task inference for language models.
- Explanations that are tuned using a validation set are especially effective, but even untuned explanations have modest positive effects and outperform carefully matched control conditions.
- However, in our experiments this capability only emerges in the largest models (although it is possible smaller models might benefit from more explanations, or on simpler tasks).

Thank You!

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